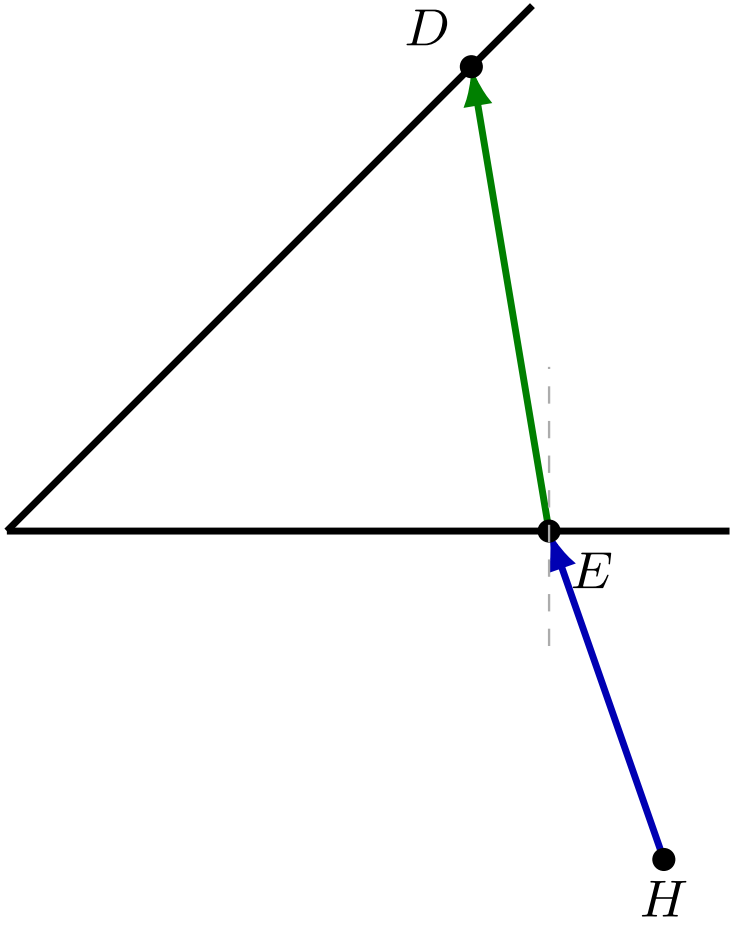
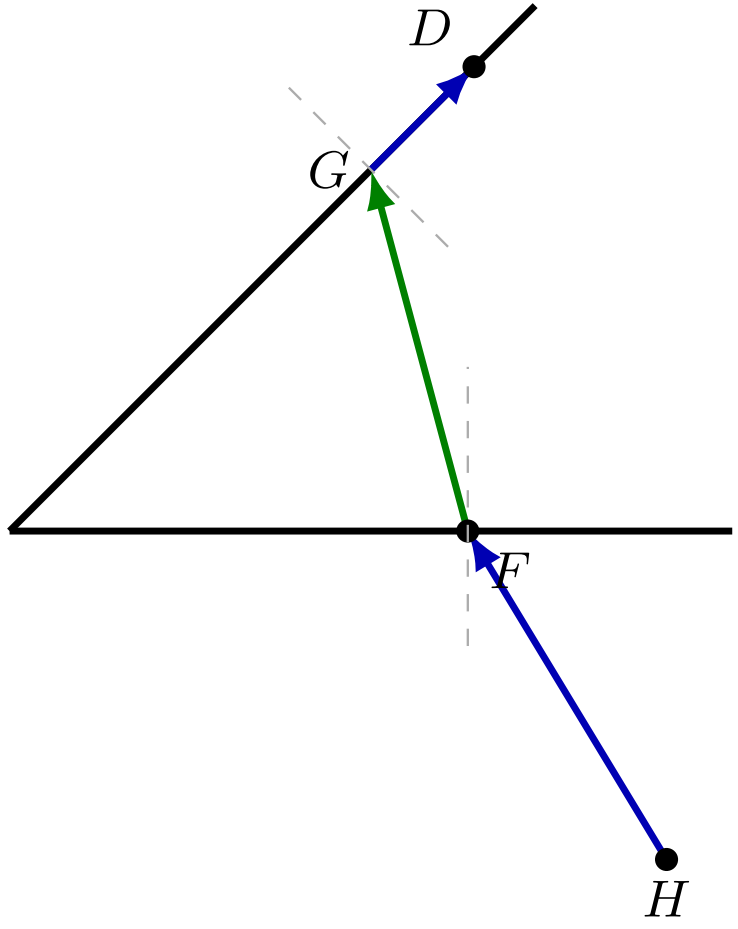


$OE = .82533$ km

$t_2 = 1963.475$ s



$OF = .69750$ km, $OG = .77796$ km

$t_3 = 1945.438$ s